

Lead Demonstrator:

Partner(s):

Period:

Date:

THE MISSION BRIEFING: You are a Lead Science Demonstrator. Your mission is to plan, engineer, and defend a live, repeatable physical demonstration that vividly proves **Newton's First Law of Motion**. You cannot simply recite the law—your demonstration must make invisible inertia visible, dramatic, and scientifically undeniable to your audience.

PHASE 1

Select Your Inertia Regime & Phenomenon

Choose **one** of the three physical categories of Newton's First Law to demonstrate. Check the box and write your demonstration title:

☐ **1. Inertia of Rest**
A stationary mass stubbornly stays at rest when a supporting surface or barrier is accelerated out from underneath it.
Idea Sparks: Tablecloth slip, coin-on-card beaker snap, embroidery hoop hex-nut drop, stacked block bottom-whip, pencil-on-ruler flick.

☐ **2. Inertia of Motion**
A moving object stubbornly continues forward in a straight line at constant velocity when its carrier abruptly halts.
Idea Sparks: Cart passenger obstacle barrier crash, skateboard curb stop, water inertia slosh cart, rolling marble barrier gap launch.

☐ **3. Inertia of Direction**
An object in circular motion flies off along an exact, straight tangent line vector the instant the centripetal inward force is cut.
Idea Sparks: Whirling string cutter, curved pie-pan marble exit gap, spinning bicycle wheel water throw, tetherball release.

Demonstration Title / Codename:

PHASE 2

Bill of Materials & Apparatus Inventory

COMPONENT ITEM	PHYSICS ROLE IN DEMONSTRATION	SOURCE (CLASSROOM/HOME)	CHECK / VERIFIED
1. The Inertial Mass:	Target object that resists acceleration (large mass = high inertia)		[] Ready
2. The Support / Carrier:	Surface or vehicle accelerated out from under or halted		[] Ready
3. The Force Device:	Applies the rapid external impulse (pull, strike, stop, cut)		[] Ready
4. Safety Catch / Basin:	Cushion, container, or barrier to contain moving items		[] Ready

Safety Rule: No glass breakables or dangerous projectiles. Design a soft catch landing zone!

Inertia Maximizer: High mass (kg) + low friction (smooth surfaces) = clear proof!

PHASE 3

Annotated Blueprint & Setup Diagram

☒ Draw apparatus BEFORE action

☒ Label Inertial Mass & Support Surface

☒ Draw arrow showing Applied Force direction

☒ Draw dashed arrow showing final path

Grid spacing: 14 mm · Draw neat, labeled vectors with a ruler

PHASE 4

Execution Protocol (Step-by-Step Procedure)

1. Setup & Baseline Alignment
How is the apparatus positioned before the demonstration begins?

3. Audience Observation (The Result)
What does the carrier do vs. what does the inertial mass do?

2. The Action (Quick Impulse Trigger)
What exact physical action do you perform? (Speed, angle, snap)

4. Repeatability & Troubleshooting
What common mistake causes this to fail? How do you prevent it?

Unit 2: Dynamics · Newton's First Law Demonstration Planner

Page 1 of 2 (Turn over for Physics Defense & Real-World Connection)

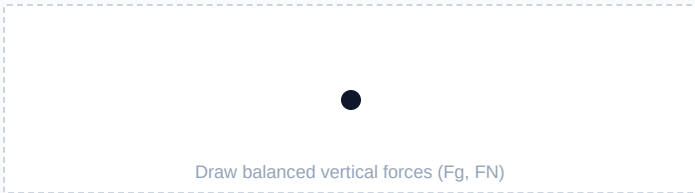
rrmudry.github.io/physics

PHASE 5Free-Body Diagram (FBD) Defense: Proving Net Force

Construct isolated Free-Body Diagrams for the **Inertial Mass** during the two critical phases. Draw labeled force vectors with proper relative arrow lengths:

Diagram A: Initial State (At Rest or Uniform Motion)

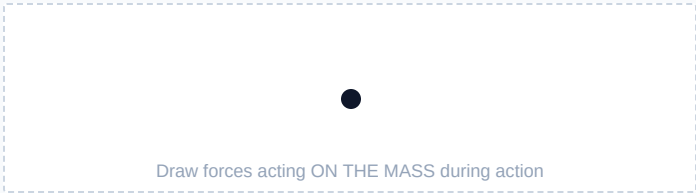
$\Sigma F = 0$
(Equilibrium)



Vertical Balance: $F_N = F_g$ (Normal force cancels gravity)
Horizontal State: $F_{\text{net}} = 0 \text{ N} \rightarrow$ Velocity is constant ($v = 0$ or v_0).

Diagram B: During Split-Second Action

Horizontal Force $\approx 0 \text{ N}$



Key Misconception Buster: Inertia is a *property of mass*, NOT a force! There is ZERO forward force pushing the mass.

PHASE 6Scientific Defense: Claim - Evidence - Reasoning (CER)

1. CLAIM:State clearly how this demonstration proves Newton's First Law.

2. EVIDENCE:Describe the exact physical motion observed during your test trials.

3. REASONING:Explain the physics using core terms: **Mass**, **Inertia**, **Net External Force (F_{net})**, and **Friction (Static vs. Kinetic)**.

PHASE 7Everyday Life & Modern Engineering Connection

Newton's First Law governs human survival and engineering every day. Connect your demonstration category to an authentic real-world technology or daily experience:

A. Real-World System or Vehicle:
(e.g., Automotive seatbelt lock, airbag sensor, earthquake base-isolator, ketchup bottle tap, bicycle mud flap)

B. How Inertia Operates in this System:
What object stubbornly wants to stay at rest, keep moving, or travel straight?

C. Engineering Solution or Consequence:
How did engineers design around this inertia to protect human life or accomplish work?

D. What Happens if Newton's First Law is Ignored?
What injury, mechanical failure, or disaster occurs without an external restraining force?

PHASE 8 Demonstration Defense & Scoring Rubric			
Criteria	Exemplary (4 pts)	Proficient (3 pts)	Developing (1–2 pts)
1. Physical Setup & Repeatability	Setup is elegant, completely safe, and successfully repeated 2+ times with clear dramatic effect.	Setup works reliably; demonstrates the intended category with minor execution stumble.	Fails on multiple attempts; apparatus unstable or pull speed too slow to overcome friction.
2. Free-Body Diagram Accuracy	Both FBDs have perfectly labeled arrows, correct relative lengths, and explicit net force notation.	FBDs are correct with minor labeling omission; correctly shows absence of "forward inertia force."	Draws unphysical forward force arrow or confuses normal force with inertia.
3. Scientific Defense (CER)	Seamlessly links evidence to Mass, Inertia, Friction, and $F_{\text{net}} = 0$; debunks common myths.	Clear explanation using mass and inertia; minor imprecision with friction threshold.	Vague explanation; describes what happened without scientific reasoning or Newton's 1st Law.
4. Real-World Synthesis	Deep, accurate connection to automotive safety or modern engineering with life-saving insight.	Clear everyday connection explained with correct inertia principles.	Superficial connection; fails to explain the physical mechanism or engineering response.

Unit 2: Dynamics · Newton's First Law Demonstration Planner

Page 2 of 2 · Self-Check & Teacher Defense Score: _____ / 16 pts

rrmudry.github.io/physics